ATOM COUNT CARD

Datu Puti: Calculating acetic acid within a vinegar



PRODUCT IDENTITY

PRODUCT

• Datu Puti, Vinegar 1L

COMPOUND

• Acetic acid, CH_3COOH (molecular formula $C_2H_4O_2$).

WHY IT MATTERS

• Acetic acid is the main component of vinegar apart from water, and it gives vinegar its sour taste and pungent smell.

STUDENT

• **Rey Ivan B. Guevara (Chemistry For Engineers).**

BASE QUANTITATIVE DATA

QUANTITY	VALUE	SOURCE	PRECISION
Net content	1L	Product label	Nominal, Not Limiting
Concentration	4.5% w/w	Kibbutz Online Grocery & Cooperative Store, n.d.	2 s.f (limiting)
Density	0.987 g/mL	Evangelista et al., 2012	3 s.f
Atomic masses C,H,O	12.011, 1.008, 15.999 g/mol	CIAAW (2024)	Not limiting
Avogadro constant, N_A	$6.022\ 140\ 76 \times 10^{23}$ mol^{-1}	BIPM (2019)	Exact
Formula $C_2H_4O_2$	Subscripts 2,4,2	PubChem (NCBI, n.d.)	Exact counts

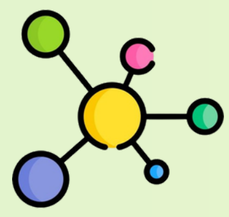
Assumptions: Acidity (4.5 %) comes from a retailer's description read as w/w, while the printed standardized label for it says its 5% research says its a rounded value; Temperature not reported.

MOLAR MASS BUILD-UP

ELEMENT	ATOMS	G/MOL (EACH)	G/MOL	% BY MASS
C	2	12.011	24.022	40.002
H	4	1.008	4.032	6.714
O	2	15.999	31.998	53.284
$C_2H_4O_2$	8		60.052	100.000

ADDITION RULE: EACH CONTRIBUTION HAS 3 DECIMAL PLACES

THE MOLE MAP: 1L bottle



1000 mL solution → (x4.5/100) → 45 mL AcOH → (x0.987 g/mL) → 4.4415×10^1 g AcOH → (÷60.052) → 0.7 mol AcOH → (xN_A) → 4.2×10^{23} molecules $C_2H_4O_2$ → (x4 H atoms per molecules) → 1.7×10^{24} molecules $C_2H_4O_2$

Exacted from mL solution to mL AcOH, rounded from g AcOH up to the final process

MOLE CONVERSION CALCULATIONS

Molar mass of $C_2H_4O_2$

2(12.011) + 4(1.008) + 2(15.999) = **60.052 g/mol**

Label bridge: volume → mass of AcOH

$$1000\ \cancel{mL\ sol} \times \frac{4.5\ mL\ AcOH}{100\ \cancel{mL\ sol}} = 45\ mL\ AcOH \times \frac{0.987\ g}{1\ mL\ AcOH}$$

= 44.415 g = **4.4415×10^1 g AcOH**

Mass → moles

$$44.415\ \cancel{g} \times \frac{1\ mol}{60.052\ \cancel{g}} = 0.7396\ mol$$

= **7.4×10^{-1} mol AcOH**

Moles → molecules

$$0.7396\ mol \times \frac{N_A\ molecules}{1\ mol} = 4.4540 \times 10^{24}$$

= **4.5×10^{24} molecules**

Atoms of one element inside the formula

$$4.4540 \times 10^{24}\ molecules \times \frac{4\ H\ atoms}{1\ molecule} = 1.7816 \times 10^{25}$$

= 1.8×10^{25} H atoms

Percent composition + two-route check (hydrogen)

$$\%H = \frac{4.032\ g/mol}{60.052\ g/mol} \times 10000\% = \mathbf{6.714\%}$$

Route A: 44.415 g x 0.06714 = **2.982 g**

$$\text{Route B: } 0.7396\ \cancel{mol\ AcOH} \times \frac{4\ mol\ H}{1\ \cancel{mol\ AcOH}} = 2.9584\ mol\ H$$

$$2.9584\ \cancel{mol\ H} \times \frac{1.008\ g}{1\ \cancel{mol\ H}} = \mathbf{2.982\ g}$$

Both routes → **2.9×10^0 g H** ✔

Reverse: from % composition to the formula

In 1000g: **C** 40.002 ÷ 12.011 = 3.3304 mol; **H** 6.714 ÷ 1.008 = 6.6607 mol; **O** 53.284 ÷ 15.999 = 3.3305 mol

÷ 3.3305 → 2.000 : 4.000 : 2000 → **$C_2H_4O_2$**

Reverse: how much solution in one mole?

$$1.00\ \cancel{mol} \times \frac{60.052\ \cancel{g}}{1\ \cancel{mol}} \times \frac{1\ mL\ \cancel{AcOH}}{0.987\ \cancel{g}} \times \frac{100\ mL\ soln}{4.5\ \cancel{mL\ AcOH}} = 1352.07\ mL$$

→ **1.3×10^3 mL solution**

In % composition of hydrogen atoms 10000% was used as an equivalent of 100

SCALE-UP: FROM ONE BOTTLE TO A RESTAURANT

ONE BOTTLE

1000 mL of 4.5% solution

0.7 mol AcOH

4.2×10^{23} molecules

ONE RESTAURANT, ONE YEAR

Assuming the restaurant consumes 50 bottles/year

3.7×10^1 mol

50 x 0.7396 mol
= 36.98 mol = 2.2 kg AcOH

ONE HUNDRED RESTAURANT, ONE YEAR

For 100 restaurants averaging the same per year

3.7×10^3 mol

5000 x 0.7396 mol = 3698 mol = 222 kg AcOH

REVERSE

One mole of AcOH is in

1.3×10^3 mL

of solution, almost ⅓ more than the bottle

SIGNIFICANT-FIGURE

• 1000 mL **- Ambiguous**
• 4.5% **- 2 s.f (numbers taken from source only contained 2 digits)**
• 0.7 mol **- 2 s.f {rounded}**
calculator = 0.7396090055 mol
• (% by mass values)
40.052 ; 6,714 ; 53.284 **{rounded to 3 s.f decimal points}**
calculator:
40.00199827
6.714181043
53.28382069
• 2.9×10^0
• 1.3×10^3 mL **- 2 s.f {rounded}**

Most result values in this atom count card infographic are rounded to utmost 4 decimal points

ONE-NUMBER TAKEAWAY

One whole 1 L bottle of Datu Puti holds about 4.5×10^{23} molecules of acetic acid, which is still less than one mole, meaning it would take about 1.3×10^3 mL of vinegar to collect just 6.022×10^{23}, which is simply a mole of acetic acid and yet acetic acid remains as the main component of a vinegar.

AI DISCLOSURE

The specific AI used for this exercise are Claude and ChatGPT wherein ChatGPT was used merely to generate the vinegar image and Claude was used for citaitions and scan them for information needed before the actual calculation and it was also used to select the acetic acid compound which has similar properties to the compound example of the exercise criteria and instruction files.

CITATIONS

•National Center for Biotechnology Information. (n.d.). PubChem compound summary for CID 176, acetic acid. Retrieved October [day], 2026, from PubChem. (2019). Acetic acid. Nih.Gov; PubChem. https://pubchem.ncbi.nlm.nih.gov/compound/Acetic-acid
•Evangelista, D. K. O., Gotuato, G. B., Ocbina, J. B., & Padua, J. A. H. (2012). Determination of total acidity of vinegar [Unpublished laboratory report]. Philippine Normal University. Studylib. File. (2016, March 10). Studylib.Net. https://studylib.net/doc/9584923/file
•Bureau International des Poids et Mesures. (2019). The International System of Units (SI) (9th ed.). SI brochure - BIPM. (2026). BIPM. https://www.bipm.org/en/publications/si-brochure
•Kibbutz Online Grocery & Cooperative Store. (n.d.). Datu Puti vinegar 1 liter. Kibbutz. Retrieved October 17, 2026, from https://kibbutz.ph/product/datu-puti-vinegar-1-liter/